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Eastern Variscan fold belt: Paleomagnetic evidence for oroclinal bending

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ABSTRACT

The Rhenohercynian zone of Variscan Europe trends from southern Portugal, around the Ibero-Armorican arc, into southern Britain, Belgium, northern Germany, and the Moravo-Silesian zone of the east Czech Republic. Although the geodynamic evolution of the central and western segments of this arcuate belt is now fairly well understood, the scenario for the more easterly parts remains enigmatic. A paleomagnetic study of Upper Devonian rocks from the Moravo-Silesian zone has been undertaken. Two ancient directions of magnetization have been identified. The first (the A magnetization) is secondary in origin and relates to a remagnetization of Permian-Carboniferous age. The B magnetization passes the fold test, is considered Late Devonian in age, and agrees in terms of inclination to previously published paleomagnetic results from coeval rocks of the Harz Mountains and Franconian Forest, central Europe. The declination, however, differs by about 90°. Restoring the Moravo-Silesian zone to its original Devonian orientation as defined by the new paleomagnetic data brings the structural trends of central and eastern Europe into agreement. Because they were previously part of the same tectonostratigraphic unit, as suggested by the stratigraphic record, this implies that rotational oroclinal bending around the northern and eastern flanks of the Bohemian massif occurred sometime between the Late Devonian and Late Carboniferous.

INTRODUCTION

Several models have been proposed to explain the arcuate shape of the Variscan fold belt in Europe. With regard to the Ibero-Armorican arc, Carey (1955) originally postulated secondary processes with bending of an originally straight belt. It has since been demonstrated, however, that this arc was to some extent primary in nature, but underwent secondary tightening during the final stages of the Variscan orogeny (Julivert and Arboleya, 1986; Pérez-Estaún et al., 1988). This is supported by the paleomagnetic record, which shows a close correlation between the declination deviations and the structural trend. It is not, however, a 1:1 correlation as would be expected for purely secondary (oroclinal) bending (Bonhommet et al., 1981; Eldredge et al., 1985; Hirt et al., 1992; Lowrie and Hirt, 1986; Perroud, 1986; Ries et al., 1980; Ries and Shackleton, 1976; Bachtadse and Van der Voo, 1986). Various mechanisms for the cause of bending throughout the fold belt have been suggested, ranging from strike-slip models (Badham, 1982), and large-scale shear zones and oblique collision (Bachtadse et al., 1983), to draping of the more mobile regions around relatively stable blocks (Ziegler, 1990), or the accretion of microplates onto irregularly shaped continental margins (Lorenz, 1976). Matte and Ribiero (1975), Burg et al. (1981), and Matte and Burg (1981) suggested that the arcuate shape of the Ibero-Armorican arc was caused by indentation of a southern continental promontory, or microplate (Ziegler, 1990; Riding, 1974), into the northern continents. This model is supported by paleomagnetic evidence (Hirt et al., 1992; Parés et al., 1994).

The situation for the more eastern parts of the belt is not so clear. Correlation of the Rhenohercynian with the Moravo-Silesian zone southeast of the Bohemian massif has been made on the basis of structural and stratigraphic evidence. It has been proposed that they were formerly connected by an arcuate structure around the eastern flank of the Bohemian massif, which has subsequently been obliterated by faulting along the southwest margin of the Eastern European platform (Burchette, 1981; Franke, 1989). However, whether the arc was primary in origin, or secondary and resulting from oroclinal bending of the Rhenohercynian around the Bohemian massif, is not known.

In order to establish the structural relationship between these two regions, and to test for oroclinal bending in the eastern sector of the Variscan fold belt, a paleomagnetic study of Devonian sedimentary and volcanic rocks from Moravia has been carried out.

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The Moravo-Silesian zone is separated to the west from the Moldanubian zone by a northwest-dipping, dextral transpressional zone of strike slip (Schulmann et al., 1991). This shear zone was active in late Paleozoic time and probably reflects reactivation of an older southeast-facing thrust (Matte, 1991). Basement rocks of Moravo-Silesian are composed of Cadomian granitoids (e.g., the Brno pluton) that have late Precambrian–Early Cambrian (550–580 Ma) intrusion ages (Scharbert and Batik, 1980; Sorauf and Jäger, 1982; van Breemen et al., 1982). These are overlain with an erosional unconformity by thin epicontinental Middle Devonian red sandstones, conglomerates, spilitic submarine volcanic rocks, and Upper Devonian–Lower Carboniferous reef carbonates of the Moravian Karst (Fig. 1).

Deformation in the region increases toward the northwestern margin with the Moldanubian, and also along strike to the northeast, such that to the southwest the basement and cover rocks are only weakly deformed, whereas to the northeast they are strongly sheared with northeast-southwest lineation and northwest-dipping foliation. This deformation is related to post-Devonian northeast-directed thrusting of the Moldanubian over Moravia (Fritz and Neubauer, 1995; Matte et al., 1990). It has been suggested that this oblique collision and dextral transpressive regime, which is also recorded in the structural record of Devonian sequences along the northwest Moravian margin (Rajlich, 1989), are the result of indentation of the Moldanubian with Moravo-Silesia (Fritz and Neubauer, 1991). The Carboniferous foredeep has also been strongly deformed with upright northeast-southwest–trending folds with steeply dipping cleavage and horizontal lineation fabrics related to transpressional dextral tectonics. Dextral transpressional movement along the Moldanubian boundary is determined stratigraphically as post-Namurian and pre-Permian, and by $^{40}Ar/^{39}Ar$ plateau ages of 305–310 Ma from associated gneisses (Matte et al., 1990; see also Dallmeyer et al., 1995).

SAMPLING

Because of limited outcrop conditions and the relatively high degree of deformation and metamorphism prevalent in the Moravo-Silesian zone, sampling was restricted to seven separate localities (Fig. 1). The samples comprise Middle Devonian clastic sediments (BRN and ZNO); Givetian-Frasnian Josefov reef limestones (JED)

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containing rich brachiopod faunas, abundant corals, and stromatoporids; upper Famennian Krtiny limestones (KRT), comprising gray and reddish-gray condensed nodular carbonates; and Famennian fine-grained, dark gray, fossiliferous carbonates (HAD). Mid-Upper Devonian spilitic submarine volcanic rocks were sampled at two localities farther to the northeast (CHA and OND).

PALEOMAGNETIC RESULTS

All samples were thermally demagnetized, and results were analyzed using orthogonal vector plots (Zijderveld, 1967) and stereographic projection. The directions of the linear segments were calculated using the principal component analysis of Kirschvink (1980), and statistics were calculated in all cases according to Fisher (1953). The magnetic mineralogy was determined using isothermal remanent magnetization (IRM) techniques, and thermal demagnetization of the saturation IRM (SIRM). Although none of the samples has undergone any apparent penetrative deformation, the anisotropy of magnetic susceptibility was measured on a number of samples from each rock type using a KLY-2 Kappabridge. The results show that, for all rock types, only very low degrees of magnetic grain alignment (generally 1%–2%) are present, indicating that no secondary fabrics have been impressed on the rocks.

Reliable magnetization directions could only be identified from the sedimentary samples. During thermal demagnetization, the volcanic rocks were magnetically unstable and failed to reveal any consistent magnetization direction at site level. These samples were collected from the northeastern part of the Moravian zone where deformation and metasomatic activity are more pronounced, and these results are not discussed.

The sandstone and carbonate samples show distinct behavior during thermal treatment. After removal of a low-temperature (<200 °C) component that is generally similar to the local present-day field direction, up to two directions of magnetization, A and B, can be isolated. The Middle Devonian sandstones show fairly stable behavior with removal of a shallow southerly A direction of magnetization between 200 and 580 °C, during which the intensity of magnetization remains relatively constant, and the demagnetization trajectory often follows a great circle path (Fig. 2A). Above 580 °C, a high-temperature component (magnetization B) can be identified that is of mixed polarity after bedding correction. IRM experiments and subsequent thermal demagnetization of the saturation magnetization confirm the predominance of hematite as the magnetic carrier.

The carbonate samples can be subdivided into two subgroups according to their demagnetization behavior and lithology. The fine-grained gray Hady limestones belong to the first category, in which both the A and B magnetizations can be clearly identified after removal of a present-day overprint (Fig. 2B). Magnetization A is southerly and shallow and is removed between 200 and 350 °C, whereas magnetization B represents the stable end-point direction. It has unblocking temperatures of 500–530 °C and is directed toward the east with intermediate negative and positive in situ inclination values. The interpretation that this component is carried by magnetite is supported by IRM and thermal demagnetization of SIRM experiments. These show the presence of only one magnetic carrier that saturates in low fields (<100 mT) and has unblocking temperatures of 500–580 °C.

The gray, pinkish-gray, and red Krtiny and Josefov limestones show different behavior. They generally have stronger NRM intensities than the Hady Berg carbonates, but the A magnetization dominates the signal. In most Krtiny samples, the A magnetization represents the stable end point, but in some of the red samples the demagnetization trajectories are not directed toward the origin but are slightly curved, indicating the presence of a second, higher temperature (>550 °C) component of magnetization. However, the intensity of magnetization at these temperatures is very low, and the great circles are not defined well enough for interpretation. For the red and pinkish Josefov limestones, the A magnetization is dominant, and demagnetization trajectories follow a great circle path at higher temperatures. Stable end points for the B component can be

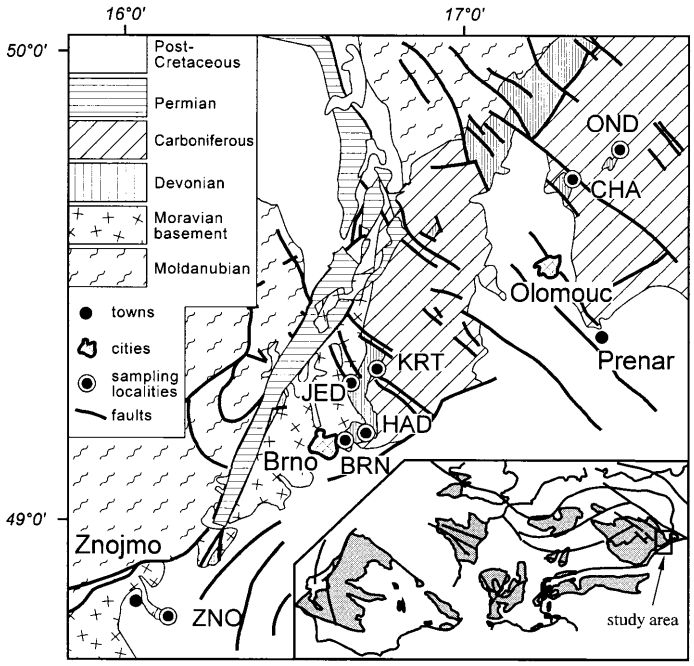


Figure 1. Geologic sketch map of sampling localities in Moravo-Silesian zone. See text for sample abbreviations.

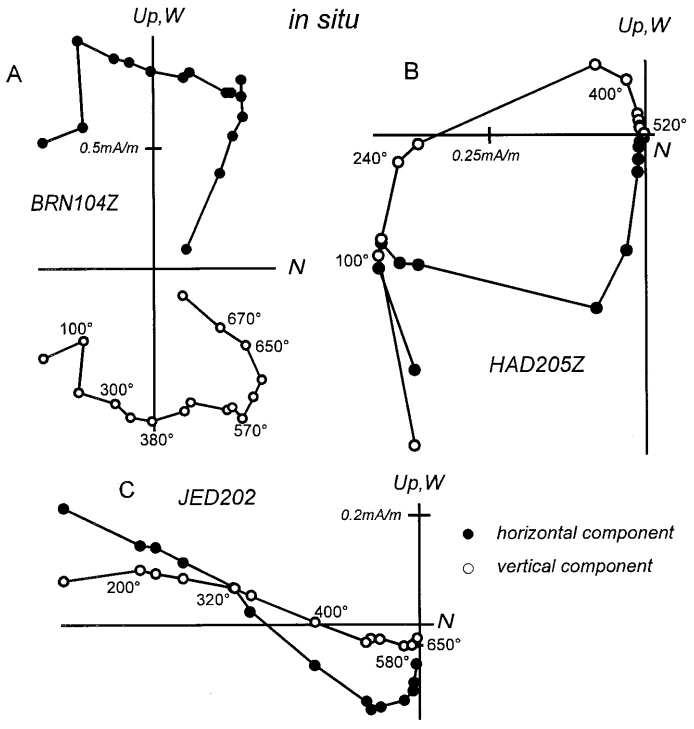


Figure 2. Typical demagnetization characteristics of Devonian sandstones (A), Hady Berg limestones (B), and Josefov limestones (C). White circles represent vertical projections, black circles represent horizontal projections.

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occasionally identified (Fig. 2C), but the great circles could not be reliably used for calculation of a site mean. Thus the mean for site JED is based on only four samples, and the corresponding α_{95} is large (Table 1B). However, the bedding-corrected site mean direction is similar to those obtained from the other five sites, qualifying it for inclusion in Table 1 and calculation of the overall mean.

IRM acquisition and subsequent thermal demagnetization experiments of samples from both KRT and JED show the presence of both magnetite and hematite as potential magnetic carriers in the red and pink samples, whereas magnetite is the dominant magnetic mineral in the gray limestones. This supports the evidence from the NRM demagnetization characteristics that in the Krtiny and Josefov limestones, magnetite is carrying the lower temperature A magnetization, and hematite is the carrier of B.

INTERPRETATION

In addition to a present-day overprint, two ancient directions of magnetization can be identified in these Middle to Upper Devonian rocks. The A magnetization is of single polarity, and yields an overall mean direction of 202°/03°, k = 12.8, α_{95} = 19.5° (six sites) in situ, and 195°/10°, k = 6.5, α_{95} = 28.3° after bedding correction (Table 1). It fails the fold test, demonstrating that acquisition postdates deformation and is therefore of secondary origin.

Deformation continued in the eastern Moravo-Silesian zone until Upper Carboniferous time, as demonstrated by the highly deformed sediments of the Carboniferous foredeep, and the final transpressional movements along the Moldanubian-Moravian boundary to the west occurred sometime between the Namurian and Stephanian. This implies, therefore, a Late Carboniferous age of deformation. Comparing the paleopole position of 42°S, 354°E derived from A with the Carboniferous-Permian sector of the European apparent polar wander path (Van der Voo, 1993), it clearly coincides with the Upper Carboniferous segment and is thus interpreted as reflecting an overprint event of this age.

The B magnetization is generally isolated as straight-line segments directed toward the origin in orthogonal projection. Identified in six sites, it is of mixed polarity, and yields an overall mean direction of 091°/−08°, k = 8.8, α_{95} = 23.9° in situ, and 089°/−37°, k = 17.9, α_{95} = 16.3° after bedding correction (Table 1B, Fig. 3). The improvement in clustering of the site mean directions after bedding correction is significant at the 95% level of confidence using the fold test of McFadden (1990). Because folding was initiated in lowermost Carboniferous time in this region, a close relationship between the magnetization and rock ages is implied, and the B magnetization is thus considered primary in origin.

However, it should be pointed out that the site mean directions have large α_{95} values, as does the overall mean direction, and the reversal test of McFadden and McElhinny (1990) is inconclusive. This is related to the strong overprint directions making isolation of the primary direction of magnetization difficult. Furthermore, the sequences studied span a large time period (Middle to uppermost Devonian), and minor relative rotations may have occurred between

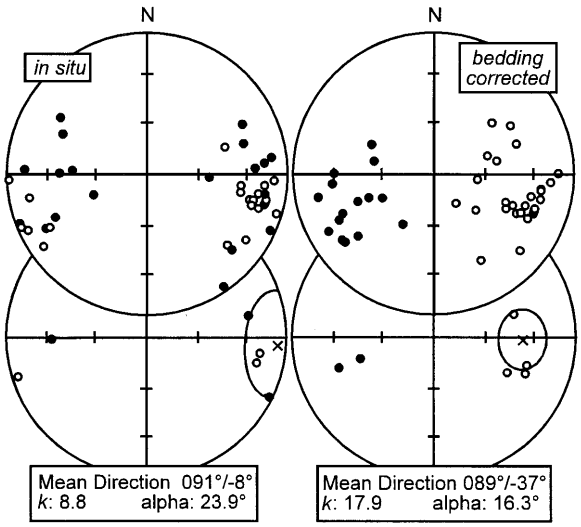


Figure 3. Stereographic projections of sample (top) and site mean (bottom) B magnetization directions before and after bedding correction. White circles represent upper hemisphere projection; black circles represent lower hemisphere projection; *k* is precision parameter (after Fisher, 1953), and α_{95} is semiangle of cone of 95% confidence.

TABLE 1A. SITE MEAN DIRECTIONS FOR COMPONENT A

Site	Lithology	*N*/*N**	In situ			Bedding corrected		
			Dec/Inc	*k*	α_{95}°	Dec/Inc	*k*	α_{95}°
BRN	sandstone	16/14	196/−02	20.6	9.0	199/−15	20.6	9.0
ZNO	sandstone	14/9	209/−02	34.6	8.9	204/+24	37.9	8.5
HAD1	limestone	8/4	200/−15	22.4	19.8	201/−12	22.4	19.8
HAD2	limestone	8/2	182/+25	-	-	172/+39	-	-
HAD3	limestone	7/2	204/+28	-	-	162/+24	-	-
JED	limestone	20/14	217/−15	75.2	4.6	225/−05	63.1	5.0
mean		6 sites	202/+03	13.4	19.0	196/+12	6.2	29.3

TABLE 1B. SITE MEAN DIRECTIONS FOR COMPONENT B.

Site	*N*/*N**	In situ			Bedding corrected		
		Dec/Inc	*k*	α_{95}°	Dec/Inc	*k*	α_{95}°
BRN	16/10	269/+33	12.3	14.4	252/+45	12.3	14.4
ZNO	14/9	246/−06	25.3	10.4	245/+29	25.3	10.4
HAD1	8/5	099/−20	61.7	9.8	107/−32	61.7	9.8
HAD2	8/6	103/−21	127.5	6.0	112/−31	127.5	6.0
HAD3	7/5	078/+27	18.7	18.1	075/−41	18.7	18.1
JED	20/4	111/−04	13.4	26.1	109/−46	13.4	26.1
mean	6 sites	091/−08	8.8	23.9	089/−37	17.9	16.3

N, number of samples measured: *N**, number of samples used in calculation of site mean: Dec/Inc, declination/inclination (in degrees): *k*, precision parameter (after Fisher 1953): α_{95}, semi-angle of the cone of 95% confidence.

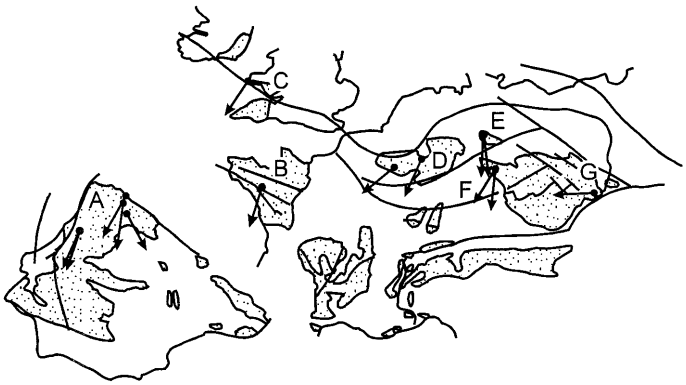


Figure 4. Sketch map of European Variscides, showing Paleozoic massifs, major faults, and observed Upper Devonian and Carboniferous declination directions (arrows). These data are derived from previously published compilations of paleomagnetic data and have been grouped into overall mean directions according to age of magnetization and general strike of regions involved. A—Iberian massif (Parés et al., 1994); B—Armorican massif (Perroud, 1986); C—Wales (Stearns and Van der Voo, 1987); D—Ardennes (Thominski et al., 1993; Edel and Coulon, 1987; Nowaczyk and Bleil, 1985); E—Harz Mountains (Bachtadse et al., 1983); F—Franconian Forest (Bachtadse et al., 1983); G—Moravia, this study. Note that paleomagnetic directions are all given in reversed polarity.

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sampling sites. Despite these limitations, the results are considered significant and provide important information with regard to the geodynamic evolution of the Moravo-Silesian zone. Krs and Pruner (1995), in a compilation of paleomagnetic results from throughout Variscan Europe, presented new data from mid-Upper Devonian sequences in the Moravo-Silesian zone. The primary magnetizations identified are also dominated by an overprint direction of Permian-Carboniferous age, revealing a poorly grouped eastern and negative high unblocking temperature component, which Krs and Pruner interpreted as primary in origin. The overall mean direction of 117°/−25°, $\alpha_{95} = 23.5°$, $k = 28.6$ (three sites, after bedding correction) is similar to that direction obtained in this study (see Table 1B), thus lending further support to the interpretation of our results.

DISCUSSION AND CONCLUSIONS

Inverting the overall mean direction for component B into its reversed polarity (269°/+37°) allows direct comparison with previously published paleomagnetic data from other Variscan massifs in central Europe (Fig. 4). Within the error limits, the inclination value for B is in reasonable agreement with the data of Bachtadse et al. (1983). However, comparing the declination values, it is clear that the Moravo-Silesian zone has rotated ~90° clockwise since the Late Devonian, with respect to the German segments of the belt. Anticlockwise rotation of the Moravo-Silesian zone by 90° brings both the paleomagnetic declination and structural trend into agreement with those of the Rhenohercynian zone of central Europe. Because they were formerly part of the same tectonostratigraphic unit, as indicated in the geologic record, this implies that rotational oroclinal bending has occurred around the northeast flank of the Bohemian massif. The timing of this rotation can be bracketed by component A, which was acquired in the Upper Carboniferous and is not rotated. The mechanism responsible for bending remains unclear, but a likely contender would be that of indentation of the Bohemian massif into the Rhenohercynian.

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